

Exposure and chain attachment for an impacted tooth

A permanent tooth is still beneath the gum or within the jawbone. Surgical exposure and a small bonded chain can make it accessible for orthodontic alignment. The aim is to keep your own tooth.

Understanding an impacted or displaced tooth

An **impacted tooth** is present but has not erupted normally. **Displacement** means that its position or direction differs from the normal eruption path. Different teeth in the upper or lower jaw can be affected.

Tooth position and surroundings

We assess the tooth's position, direction and root shape, its relationship to neighbouring roots and other anatomical structures, and the gum overlying it.

Space and planned movement

The tooth needs sufficient space and a suitable path of movement. Together with the orthodontist, we plan access, attachment and the subsequent alignment.

Examination and joint planning

We assess your mouth, existing X-rays and the orthodontic treatment plan. If an important question remains unanswered, a CBCT scan for three-dimensional planning may help, for example where the relationship to adjacent roots or the nerve canal is unclear. It is not required for every impacted tooth.

Before surgery, we agree whether space must first be created, where the chain will be secured and who will continue the tooth movement. Exposure alone does not replace orthodontic treatment.

Other treatment options

Depending on position, age and root development, options include monitoring at agreed intervals, removing an obstruction or creating space orthodontically. Sometimes exposure without an immediate chain is sufficient. If the outlook is poor, removal with a plan for the resulting space may be more appropriate. Surgical uprighting or tooth transplantation may be options in selected cases and require a separate discussion.

Without treatment, the tooth may remain unchanged within the jaw. However, adjacent roots may be damaged, infection or a cyst may develop, or a gap may remain. Whether monitoring is reasonable depends on your individual findings.

How exposure and chain attachment work

1. **Anaesthesia and access.** Once the local anaesthetic is effective, we open the gum. If necessary, we remove a small amount of bone over the crown.
2. **Bonding the attachment and chain.** A small attachment with a fine chain is bonded to the clean, dry tooth surface. The attachment is on the crown; no loop is placed around the root.
3. **Managing the wound.** With closed exposure, the gum is replaced and stitched, leaving the secured chain end accessible. With open exposure, planned access to the crown remains, sometimes protected by a dressing.
4. **Orthodontic movement.** Controlled traction or uprighting starts at the agreed time. Your orthodontist sets the direction and force.

Open the image sequence showing exposure and chain attachment

The approach depends on the tooth position and the gum. A baby tooth, extra tooth or another tooth is removed only if this has been separately planned and discussed.

Possible risks and limitations

- **Symptoms and healing:** pain, swelling, bruising, bleeding and limited mouth opening; infection or delayed wound healing may also occur.
- **Teeth and gums:** damage to neighbouring teeth, roots, fillings or brace components; gum recession, scarring, pockets or differences in the gum line.
- **Attachment and chain:** loosening, breakage, overgrowth by gum tissue or loss of access. Rebonding or repeat exposure may be necessary. Very rarely, detached parts are swallowed or inhaled.
- **Tooth movement:** alignment may fail if the tooth is fused to bone (ankylosis), has an eruption disorder or lies unfavourably. Root shortening or damage to the treated or neighbouring teeth and loss of tooth vitality are possible. Further treatment or removal may be required.
- **Depending on location:** nerves in the lower jaw may be injured, causing temporary or permanent altered feeling in the lip, chin or tongue, and rarely altered taste. A tooth high in the upper jaw may be close to the nasal floor or maxillary sinus; an opening into the sinus may require closure and additional precautions.
- **Medication and anaesthesia:** intolerance, allergic reactions and circulation or breathing problems can occur; severe reactions are rare.

The wound usually heals much sooner than the tooth reaches its final position. Movement often takes many months; the complete orthodontic treatment may take longer than two years. Success cannot be guaranteed.

Anaesthesia and comfort options

Method	Description	Cost
Local anaesthesia Standard	You remain awake and feel pressure, not pain. If you feel pain, more anaesthetic is given immediately. You may use public transport home.	Statutory insurance benefit or included in the treatment package
Oral sedation (Tavor) Comfort option	A tablet about 60 minutes beforehand. It reduces stress without inducing sleep. Vital signs are monitored. An escort is needed; no driving for 24 hours. Collection from the practice is mandatory.	Approx. €80 Out-of-pocket
IV sedation Comfort option	A professional anaesthesia team provides medication at the sleep–wake boundary. You remain cooperative and usually have little memory. Recovery room and vital-sign monitoring afterwards. Tuesdays only. No food from 10 pm and no drinks from 6 am. Collection from the practice is mandatory.	Approx. €230 Out-of-pocket
General anaesthesia Comfort option	Deep sleep, with a breathing tube through the nose into the airway. No awareness of surgery. Recovery room and vital-sign monitoring afterwards. Tuesdays only. Same fasting instructions as IV sedation. Collection from the practice is mandatory.	Approx. €450 Out-of-pocket

Local anaesthesia is the standard and is sufficient for most patients. Sedation is an optional comfort service paid out of pocket. With every method, the anaesthetic is fully effective before we begin so that you do not feel pain.

For children and adolescents, the method is chosen individually according to age, health and cooperation. Tavor is not a routine recommendation for minors. Your agreed written fasting and medication instructions take priority. Medically necessary general anaesthesia and possible insurance coverage are assessed separately.

Preparing for treatment

- Bring or arrange transfer of your referral, orthodontic plan, existing images, medication list and allergy details.
- Tell us about blood thinners, diabetes, immune suppression, bisphosphonates/denosumab and a possible pregnancy. Do not stop prescribed medicines on your own.
- With local anaesthesia alone, you can normally have a light meal beforehand. For sedation or general anaesthesia, follow the agreed instructions.
- For minors, clarify parental responsibility and consent in advance; involve children and adolescents in an age-appropriate discussion.
- Arrange the check-up, stitch removal and orthodontic follow-up. After sedation/general anaesthesia, arrange collection and supervision; for at least 24 hours, do not drive, operate machinery, drink alcohol or make important decisions.

Costs: Surgical exposure, orthodontic treatment, any CBCT scan and comfort anaesthesia are considered separately. We discuss coverage and any out-of-pocket costs beforehand.

After the procedure

On the day: rest, keep your head slightly raised and cool the cheek from outside with breaks. Do not rinse vigorously, suck at or disturb the wound. Eat only when numbness has worn off; start with soft, lukewarm food and chew on the other side.

Over the next few days: resume sport and strenuous activity only as advised. Do not smoke. Continue brushing the other teeth; clean the wound area, chain and brace gently as instructed. Use mouth rinses and antibiotics only if prescribed.

Reviews: stitches are removed after 1–2 weeks. The chain remains for orthodontic treatment. A dressing is changed or removed only as instructed. We agree the start of traction with your orthodontist.

Pain relief

Children and adolescents receive a **personal plan based on age and weight**. The following is the practice's adult regimen and applies only after individual approval. Medical conditions or other medication may require lower doses or a different drug.

Medicine	Dose if suitable	Note
Ibuprofen 600 mg	Every 6–8 hours, max. 3 doses/day	Take with food
Paracetamol 500–1000 mg	Alternatively/additionally every 6 hours	Max. 4 g/day; not suitable for all body weights
Novalgin drops	Up to 30 drops 4 times/day, only if prescribed	Only at 500 mg/ml and 20 drops/ml
No additional aspirin/ASA	Avoid as a painkiller	Do not independently stop prescribed long-term aspirin

If fever, chills, sore throat or painful mucosal sores develop while taking Novalgin/metamizole: **stop it immediately and seek urgent medical assessment**. Do not add paracetamol-containing combination products without advice.

If the chain becomes loose

Do not pull, shorten, glue or tighten it yourself. Contact us and your orthodontist promptly, even without pain. Sudden coughing or breathing difficulty after a part detaches may indicate inhalation; call 112 immediately for breathing difficulty.

Contact us immediately for warning signs

Bleeding that continues despite 30 minutes of pressure with clean gauze, fever, pus or markedly increasing pain and swelling: practice **06151 786540**.

Post-operative problems outside consultation hours: **+49 151 10437450**. If unavailable: dental emergency service **01805 60 70 11**.

For a life-threatening emergency, breathing difficulty, severe swallowing difficulty or rapidly increasing swelling of the floor of the mouth, neck or tongue, **call 112 immediately**.

Frequently asked questions

Is the tooth pulled into place during the operation

No. Exposure provides access; the chain connects the tooth to later controlled orthodontic movement. Surgical uprighting is a different procedure requiring a separate plan.

Is a chain suitable for every impacted tooth?

Whether a chain, another orthodontic appliance or a different approach is appropriate depends on the individual findings. Tooth position, available space and the ability to move the tooth are important. The surgeon and orthodontist decide this together.

Is closed exposure better than open exposure

No single technique is best for every tooth. Tooth position, traction direction and protection of the gum are decisive. We explain your planned approach after examination.

Can the chain lie beneath the gum

Yes. With closed exposure, part lies under the gum. The intended end must remain accessible for further treatment. If it disappears or the area hurts, arrange a review.

What if the tooth does not move

Your orthodontist first checks the attachment, available space, traction direction and progress. Suspected ankylosis or another eruption disorder requires reassessment. Simply pulling harder is not a solution. Other treatment options may need to be discussed.

Next steps checklist

- Which tooth is being exposed, and which technique has been agreed?
- Are available space and orthodontic follow-up settled?
- Are anaesthesia, personal medication instructions and any collection arrangements clear?
- Are the check-up and stitch removal after 1–2 weeks arranged?
- Do you know whom to contact for a loose chain and have the emergency numbers?

This information supplements your personal consent discussion. Your findings and the jointly agreed surgical and orthodontic plan determine your treatment.

Clinical sources

Prepared using the practice standard and the following sources: [BAOMS: Exposure of impacted canine](#); [Cambridge University Hospitals: Exposure of impacted upper canine](#); [British Orthodontic Society: Impacted canines](#); [Padwa et al.: Surgical uprighting of mandibular second molars, 2017](#); [AWMF 083-005: Dental CBCT](#); [BfArM: Metamizole safety information 2024](#). Accessed 9 September 2026.